ChatGPT

ThriveLens: Human-Confirmed Multimodal Food Analysis and Offline-First Fitness Intelligence with Parameter-Efficient Vision-Language Adaptation

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Executive Summary

ThriveLens 2026, also identified in the project artifacts as Omni Fitness AI, is designed as an offline-capable fitness and nutrition system in which multimodal food analysis, nutrition logging, workout execution, progress tracking, grocery and gym discovery, reminders, and grounded coaching operate over a single longitudinal user record [1]. Its most significant research contribution is not simply food-image classification. Rather, it is an **evidence-preserving decision pipeline** in which a model-produced food observation remains a candidate until the user confirms the food and portion, a traceable nutrition source resolves canonical nutrient information, deterministic arithmetic creates the meal record, and subsequent recommendations cite saved user data [1]. This separation is scientifically and operationally important because a single food photograph generally cannot establish hidden ingredients, cooking oil, preparation method, exact portion mass, allergens, or medical suitability, while large multimodal models can produce inaccurate or incomplete health-related statements. WHO consequently recommends clearly bounded tasks, appropriate oversight, transparency, stakeholder involvement, and explicit attention to errors and bias in health-oriented multimodal systems. [1]

The audited application baseline is a substantial mobile interface prototype rather than a verified production system. At the pinned repository snapshot, the application contains its major screens and user-interface flows, but authentication and artificial-intelligence responses remain simulated, the food scanner uses sample outputs, and the Food-R1 training path is not integrated with the mobile application. The dependency manifest confirms an Expo SDK 57 and React Native 0.86.3 application with React 19.2.3, Drizzle ORM, `expo-sqlite`, and Supabase libraries. The audited SQL schema is not release-ready for sensitive personal information because it explicitly disables row-level security for all listed application tables and includes a `password_hash` field in its application-level users table. Supabase's current guidance instead recommends using Supabase Auth as the identity authority, referencing `auth.users` from application profile tables, and enabling row-level security on exposed tables with policies tied to the authenticated user's identity. [2]

The machine-learning experiment specified by the blueprint adapts the public `zy12123/Food-R1` vision-language checkpoint using supervised quantized low-rank adaptation. The project configuration loads the

model in 4-bit NormalFloat representation with double quantization, computes in FP16, freezes the vision components, trains rank-64 language-side LoRA adapters, uses DeepSpeed ZeRO-2 across two NVIDIA T4 GPUs, and fixes training at three epochs without early stopping [1]. These design choices are supported by the LoRA, QLoRA, and ZeRO literature: LoRA adapts a frozen model through low-rank trainable matrices; QLoRA combines LoRA with a frozen 4-bit backbone, NF4, double quantization, and paged optimization; and ZeRO partitions redundant training state in distributed training. [3]

A central finding of the present research audit is that the project's **split counts are internally consistent but its source counts are not yet fully reconciled**. The blueprint gives 242,756 training records, 6,460 validation records, and 52,941 test records, which sum exactly to 302,157. However, the listed source counts —107,794 Open Images records, 183,893 Fruits-360 records, and 3,490 Nutrition5K records—sum to 295,177, leaving **6,980 records, or 2.31% of the declared dataset, without source attribution in the blueprint** [1]. Publication-quality results therefore require a frozen manifest that reconciles those records and resolves the blueprint's two competing archive SHA-256 hashes before training.

No ThriveLens QLoRA accuracy, nutrient error, calibration, latency, GPU-throughput, security-test, native-device, or end-to-end acceptance result is reported as measured in this paper because the supplied blueprint explicitly records that the Food-R1 notebook did not contain saved execution outputs at the time of audit [1]. Fabricating those values would invalidate the project's evidence boundary. The results section therefore distinguishes three classes of evidence: verified repository and dataset-audit findings; published Food-R1 benchmark results from the original Food-R1 investigators; and pre-registered ThriveLens result tables whose values remain measurement-gated until the prescribed experiment is executed.

The resulting research design uses the 52,941-record project test set as an untouched final evaluation set, performs model selection only through the training and validation data, uses group-aware stratification to prevent related images from crossing experimental partitions, and proposes training-only group cross-validation for computationally affordable hyperparameter screening. Group-aware cross-validation is particularly important when datasets contain multiple related images of the same underlying acquisition or object; `StratifiedGroupKFold` explicitly combines class-balance objectives with non-overlapping groups. [4] Final metrics encompass exact match, token F1, ingredient precision/recall/F1, structured-output validity, numeric MAE/RMSE/MAPE, calibration, abstention coverage, provider-confirmed nutrient error, offline persistence, duplicate-write rate, cross-account isolation, and end-to-end task completion.

The completed study should therefore evaluate ThriveLens as a **socio-technical pipeline**, not just a neural network. Its strongest defensible claim is expected to be that food recognition, uncertainty, human confirmation, nutrition-source provenance, deterministic computation, offline persistence, and constrained downstream actions can be integrated so that model-generated observations do not silently become nutrition facts. That claim becomes publishable only when the frozen data manifest, GPU training run, untouched held-out evaluation, secure serving contract, authenticated database, native-device tests, and complete daily-loop acceptance tests produce reproducible evidence.

Abstract and Keywords

Abstract— ThriveLens is an offline-capable fitness and nutrition platform designed to integrate multimodal food analysis, nutrition logging, exercise tracking, progress monitoring, and grounded coaching through a single longitudinal user record. The study investigates whether a vision-language food model can be adapted efficiently while preserving uncertainty, provenance, human confirmation, and deterministic

nutrition calculations. The machine-learning design adapts Food-R1 through supervised quantized low-rank adaptation on a declared dataset of 302157 records, using a fixed three-epoch dual-GPU protocol with four-bit quantization, frozen vision components, language-side low-rank adapters, distributed optimization, resumable checkpoints, and structured held-out evaluation. The system architecture separates food recognition from nutritional truth: image analysis generates candidates, users confirm the food and portion, external catalogues supply traceable nutrient data, and deterministic calculations create persisted meal records. Evaluation is defined at both model and product levels and includes structured-output validity, exact match, ingredient F1, numeric nutrient error, calibration, abstention, offline recovery, synchronization correctness, account isolation, and end-to-end task completion. Audit of the supplied research blueprint identifies an unresolved source-lineage difference of 6980 records and two competing archive hashes that must be reconciled before publication. The audited mobile repository also remains a prototype with simulated artificial-intelligence and authentication behavior and a database schema whose row-level security is disabled. Consequently, project-specific performance values are treated as measurement-gated rather than inferred. The proposed methodology provides a reproducible path from the verified prototype to an experimentally defensible human-confirmed multimodal fitness and nutrition system.

Keywords— multimodal learning, food recognition, nutrition estimation, vision-language model, parameter-efficient fine-tuning, quantized low-rank adaptation, human-in-the-loop artificial intelligence, offline-first mobile health, reproducible machine learning, uncertainty calibration.

Introduction and Related Work

Problem context and research motivation. Digital fitness and nutrition tools frequently treat food recognition, nutritional databases, exercise planning, behavioral history, and artificial-intelligence recommendations as separate functions. ThriveLens instead defines these elements as stages of one evidence chain: capture or manual input produces a candidate observation; the observation is validated; the validated record persists locally and, when connectivity permits, synchronizes with cloud services; saved history is then used to generate progress summaries and grounded recommendations [1]. The blueprint intentionally limits the role of computer vision: recognition is evidence about what may be visible, rather than direct authority over a user's nutritional intake.

This distinction is supported by the food-computing literature. Food analysis involves more than dish classification because mixed meals create ingredient ambiguity, occlusion, visual variation from cooking methods, and uncertainty about quantity. The Food-R1 authors similarly characterize food understanding as a family of tasks encompassing classification, ingredient recognition, recipe generation, nutrition estimation, calorie estimation, and dietary advice, and note that complex dishes challenge purely visual recognition. [5] Nutrition5K was developed partly because conventional food datasets lacked the fine-grained weight and nutrient annotations required for nutritional understanding; it includes approximately 5,000 real dishes with RGB/video information, depth information, component weights, and nutrient annotations. [6]

Operational definition of the ThriveLens framework. For this study, the ThriveLens framework is defined by the supplied ThriveLens 2026 Research Blueprint [1] together with the repository snapshot pinned by that blueprint. The repository commit was created on September 8, 2026 and is identified by SHA `244b4bbe79b4e563df94af13affbe552d2f189b0`. This evidence boundary matters because the

blueprint distinguishes observed implementation from target architecture and from measurements that are permissible only after experiments have actually run [1].

The repository describes an Expo and React Native application containing authentication interfaces, onboarding, home, workout, tracking, nutrition, mapping, coaching, food scanning, active-workout, exercise-library, workout-completion, and profile experiences. It simultaneously records that authentication is simulated, artificial-intelligence/chat output is simulated, the food scanner returns hardcoded examples, the machine-learning training code has not been integrated into the application, and the cloud/backend path is incomplete. These facts establish the correct baseline for the paper: ThriveLens is not scientifically evaluated by screenshots or screen count; evaluation must follow persistence, security, model, serving, and end-to-end contracts.

Food-R1 and the food vision-language literature. Food-R1 was introduced in 2026 as a unified multi-task food vision-language model. Its investigators constructed CalorieBench-80K and trained Food-R1 from Qwen3-VL-8B through a two-stage procedure consisting of multi-task supervised fine-tuning with chain-of-thought supervision and subsequent reinforcement fine-tuning using Group Relative Policy Optimization. [7] The original paper reports that its supervised phase used full-parameter fine-tuning for three epochs with learning rate 4×10^{-5} and global batch size 128, followed by one epoch of GRPO at learning rate 1×10^{-6}. [5]

That original Food-R1 recipe must not be conflated with the ThriveLens experiment. ThriveLens starts from the released Food-R1 checkpoint and proposes **another supervised adaptation**, using QLoRA rather than recreating Food-R1's original full-parameter SFT-plus-GRPO training [1]. The public Food-R1 project confirms that Food-R1 is trained across food datasets including Food-101, VIREO Food-172, Recipe1M, Nutrition5K, FoodDialogues, and MM-Food-100K and releases separate SFT, GRPO, and evaluation procedures. [8] The public checkpoint is identified as a Qwen3-VL model and its repository contains four BF16 Safetensor weight shards totaling approximately 17.6 GB. [9]

Qwen3-VL itself is a general vision-language architecture designed for multimodal inputs including text, images, and video. The technical report includes dense variants such as an 8-billion-parameter model, making Qwen3-VL-8B the relevant scale for interpreting Food-R1's compute requirements. [10]

Published Food-R1 results establish a useful external reference point but are **not ThriveLens results**. On CalorieBench-80K, the Food-R1 paper reports a calorie MAE of 42.56, RMSE of 112.89, and R^2 of 0.81, compared with an MAE of 195.18 for the underlying Qwen3-VL-8B baseline and 57.99 for FoodLMM-s1. [5] It also reports 95.21% top-1 accuracy on Food-101, ingredient-recognition IoU of 78.50 and F1 of 84.61 on VireoFood-172, and a Nutrition5K calorie MAE of 27.6 with percentage MAE of 14.4%. [5] These results justify Food-R1 as a credible starting point while providing no guarantee that an additional ThriveLens adaptation will improve or even preserve those values.

Parameter-efficient adaptation. LoRA addresses the expense of full fine-tuning by freezing pretrained weights and learning low-rank updates in selected neural-network layers. Hu et al. showed that this can sharply reduce trainable parameter count while retaining competitive adaptation performance in the studied language models. [11] QLoRA extends the idea by backpropagating into LoRA adapters through a frozen, quantized backbone and introduced NF4 quantization, double quantization, and paged optimizers as memory-saving techniques. [12] Those properties directly motivate the ThriveLens choice to adapt a large Food-R1 checkpoint on a constrained dual-T4 environment rather than update the entire model [1].

DeepSpeed ZeRO complements QLoRA at the distributed-training layer. ZeRO reduces replicated memory state by partitioning optimization state and, depending on stage, gradients and parameters across participating devices. [13] ThriveLens specifies ZeRO-2 with two T4 GPUs, thereby distributing optimizer and gradient state while retaining a fixed two-device execution contract [1].

Conventional visual baseline. MobileNetV2 remains relevant as a lightweight image-classification baseline because its inverted residual and linear bottleneck architecture was specifically designed to trade off accuracy, parameter count, operations, and mobile latency. [14] The ThriveLens repository contains a legacy MobileNetV2 training script, but repository inspection reveals an important discrepancy with the prose description in the blueprint. The pinned script loads an ImageNet-pretrained MobileNetV2, performs an 80/20 directory-based split with seed 123, freezes the backbone, applies global average pooling and dropout followed by a softmax output, trains using Adam, then unfreezes the last 30 backbone layers for an additional two epochs at 10^{-5}, and can export to TensorFlow Lite. The pinned code does **not** contain the blueprint narrative's claimed Dense-128 layer, random flip/rotation/zoom augmentation, `EarlyStopping`, `ModelCheckpoint`, or `ReduceLROnPlateau` logic. Therefore, the repository code is the defensible specification of the existing baseline; any alternative MobileNetV2 configuration must receive a distinct version and experimental identifier.

Food-101 itself contains 101,000 images across 101 food categories. [15] Consequently, a publication-quality MobileNet comparison should honor either its official partition or the same group-safe ThriveLens data partition rather than report a random split that is incomparable with the multimodal experiment.

Dataset literature and generalization. The ThriveLens dataset combines different visual regimes. Open Images V4 contains 9.2 million images collected from Flickr with large-scale image-level and object-detection annotations; its natural scene diversity differs substantially from controlled food-object imagery. [16] Fruits-360's official organization currently describes its 100-by-100-pixel collection as containing 183,893 images spanning 261 fruits, vegetables, nuts, seeds, and fungi, with acquisition involving repeated views of physical objects. [17] Nutrition5K, by contrast, contains real composed dishes and detailed nutritional measurements. [6] This heterogeneity is potentially valuable, but it creates a strong requirement for source-aware reporting and group-safe splitting.

Food2K is an additional possible external benchmark rather than part of the current ThriveLens dataset contract. Its authors introduced more than one million images across 2,000 food classes and studied transfer to recognition, retrieval, detection, and segmentation. [18] The ThriveLens paper should therefore identify Food2K only as an optional external validation or pretraining resource unless a run manifest explicitly proves that it was included [1].

Calibration and abstention. An operational nutrition assistant requires more than top-1 accuracy. Neural-network confidence can be poorly calibrated, meaning a numerical confidence value may not correspond to empirical correctness probability. Guo et al. demonstrated this issue and found temperature scaling effective in many of their experiments. [19] ThriveLens accordingly treats confidence and abstention as explicit evaluation dimensions [1]. In deployment, a candidate with insufficient confidence should be presented for manual search or correction rather than silently converted into a nutrition record.

Research objectives. Consistent with the blueprint, this paper investigates five objectives [1]: first, to evaluate structured multimodal food analysis rather than unconstrained prose; second, to establish durable offline-first persistence across restarts, connectivity interruptions, retries, and account changes; third, to

test correctness across the integrated food, workout, progress, and place workflow; fourth, to establish a reproducible dual-GPU training and serving procedure under the stated compute budget; and fifth, to provide privacy, security, safety, and operational evidence suitable for a consumer-facing system.

The corresponding research questions and hypotheses are formalized below.

Research question	Primary outcome
RQ1: How accurately does the adapted Food-R1 model identify food candidates and generate required structured outputs across the project dataset?	Exact match, token F1, JSON validity, ingredient precision/recall/F1, numeric MAE/RMSE/MAPE, source-balanced performance, calibration, abstention
RQ2: Does human confirmation plus catalogue-grounded nutrient computation reduce nutrition-record error relative to direct model output?	Food confirmation accuracy, correction rate, nutrient error, unresolved rate, completion time
RQ3: Does offline-first persistence preserve records and derived summaries correctly?	Restart recovery, mutation survival, synchronization idempotence, duplicate-write rate, cross-account isolation, known-answer totals
RQ4: Can the fixed dual-T4 protocol produce reproducible training and held-out artifacts within the available compute process?	Steps completed, throughput, GPU memory/ utilization, finite loss, checkpoint integrity, held-out coverage
RQ5: Can grounded coaching suggestions be translated into valid, user-approved actions without unsupported or duplicate writes?	Groundedness, invalid-action rejection, approval rate, unsupported-claim rate, duplicate-action rate

The pre-registered hypotheses are that QLoRA adaptation will improve structured food-analysis performance relative to the unadapted checkpoint; separating recognition from catalogue-derived nutrition will reduce nutrient-record error; operation identifiers and version-aware synchronization will prevent duplicate or lost records; and typed, source-grounded coaching actions will reduce unsupported actions relative to unconstrained generation [1]. These are hypotheses rather than results.

System Design and Methodology

System architecture. ThriveLens is designed around separation of responsibilities. The mobile presentation layer consumes typed domain state. Repositories mediate local and cloud persistence. SQLite stores responsive local state and pending mutations. Authenticated service endpoints provide inference and external-provider access. Food-R1 supplies candidate interpretations rather than authoritative nutrient values. External nutrition providers resolve canonical food data, and user-approved actions become persisted domain events [1]. Expo documents that `expo-sqlite` databases persist across application restarts, supporting the local persistence role proposed by the architecture. [20]

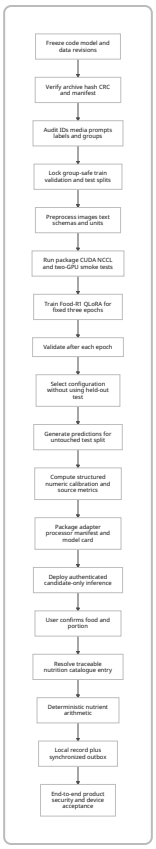


Fig. 1. Experimental and product evidence workflow for ThriveLens. Model inference intentionally precedes human confirmation and catalogue resolution rather than directly writing nutritional truth [1].

Identity and persistence contract. The target design assigns authentication to Supabase Auth and makes application profiles reference the authenticated user identifier [1]. Supabase Auth uses JSON Web Tokens and integrates with database row-level security. [21] Supabase's current security documentation explicitly states that exposed tables should have RLS enabled and describes `auth.uid()`-based policies for ensuring that authenticated users operate only on their own rows. [22] Its user-management guidance likewise recommends creating a public profile table that references `auth.users`, with RLS protecting API access. [23]

This differs materially from the audited repository SQL. The current schema creates a standalone `public.users` table including `password_hash`, and then explicitly disables RLS on users, exercises, workout plans, workout days, workout exercises, workout sessions, workout logs, food logs, progress logs, and notifications. This baseline is retained as audit evidence but excluded from the definition of a secure completed system.

For offline synchronization, each mutable record should contain a stable identifier, owner, creation and update times, record version, deletion state, and synchronization state. Each queued mutation should include a globally unique operation identifier, entity identifier, operation type, base version, payload, retry state, and timestamps [1]. Append-like events such as completed workout sets and food-log records should not be silently resolved with generic last-write-wins logic. Idempotency is tested by submitting the same operation identifier multiple times and asserting that only one domain mutation becomes visible.

Food-analysis contract. The food pipeline is:

$$\text{image or barcode} \rightarrow \text{candidate observation} \rightarrow \text{human confirmation} \rightarrow \text{catalogue record} \rightarrow \text{deterministic calculatio} \tag{1}$$

The model response must therefore include structured candidate information, uncertainty or abstention state, request identifier, and model revision, but the serving endpoint must not independently create a meal record [1]. This design prevents recognition confidence from being confused with nutrient provenance.

For packaged foods, Open Food Facts can supply barcode-oriented product information. Its official API documentation currently imposes endpoint-specific rate limits, including 15 product-read requests per minute per IP address and 10 search requests per minute per IP address, which favors explicit caching and discourages search-as-you-type designs. [24] Open Food Facts data also carry reuse obligations: the database is distributed under the Open Database License, database contents under the Database Contents License, and product images under Creative Commons Attribution ShareAlike. [25] Its API exposes completeness and data-quality information, so provider records should preserve provenance and missing-data state rather than assume completeness. [26]

For generic foods, USDA FoodData Central provides Food Search and Food Details endpoints and distributes its data under CC0. Its API requires a data.gov key, and USDA warns that the key holder must not expose that key publicly. [27] ThriveLens should therefore route credentialed provider calls through a secure service boundary rather than embed private provider credentials in the public mobile package.

Dataset design. The principal experimental dataset is specified as 302,157 records [1]. Its split counts are complete:

Split	Records	Percentage	Experimental role
Training	242,756	80.34%	Parameter updates only
Validation	6,460	2.14%	Epoch diagnostics and model selection
Test	52,941	17.52%	Untouched final generation and evaluation
Total	**302,157**	**100.00%**	Frozen experiment population

The blueprint's source declaration requires additional reconciliation.

Declared source	Project records	Share of declared total	Source characteristics relevant to evaluation
Open Images food crops	107,794	35.67%	Natural-scene imagery derived from a broad visual dataset; exact crop-generation procedure is unspecified [1]. Open Images V4 itself contains 9.2 million images and extensive object annotations. [16]
Fruits-360	183,893	60.86%	Highly repeated object imagery; the current official dataset organization reports 183,893 100-by-100 images across 261 categories. [17]

Declared source	Project records	Share of declared total	Source characteristics relevant to evaluation
Nutrition5K	3,490	1.16%	Project subset of the approximately 5,000 real dishes with nutritional annotations in Nutrition5K; project selection criteria are unspecified. [6]
Unreconciled source attribution	**6,980**	**2.31%**	Difference between declared total and sum of the three source counts; source is unspecified in the blueprint.
Total	**302,157**	**100.00%**	Split total is arithmetically consistent; source lineage is not yet complete.

The project should not reinterpret the 6,980-record difference as an additional dataset without evidence. Instead, the successful training manifest must enumerate every record and source identifier so that the 302,157 records reconcile exactly.

A second provenance issue concerns the archive hash. The blueprint records one earlier SHA-256 value,

`6c5320da2316117583e82516fa4431986b4dc36117efec8aafa9e0779e68961e`,

and a different hash expected by the notebook,

`4c378fc53d427fff808a2bfee7d923d5569355fd3794b5ffd5bfc3a4115d7f49` [1].

Neither should be selected rhetorically. The final experiment must identify the actual archive filename, byte size, member count, SHA-256 hash, manifest hash, and split files that were consumed. The other hash then remains documented as a superseded or incompatible artifact.

Dataset features and labels. The blueprint specifies a multimodal conversation-style dataset rather than a classical tabular feature matrix [1]. Required record-level fields are therefore expected to include a stable sample identifier, source, group identifier, media URI or archive member, task identifier, instruction/conversation content, image placeholder alignment, target answer, and train/validation/test assignment. Structured tasks additionally require parseable target objects; ingredient tasks require target ingredient sets; numeric tasks require field names, values, and units. Exact task-count distributions are **unspecified** and must be emitted by the frozen manifest before publication.

The exact group key used for leakage prevention is also **unspecified**. This must be resolved before training. For Fruits-360 in particular, multiple views can derive from the same physical object/acquisition process, so treating every image as independent may yield artificially optimistic results. The official dataset description documents its repeated image-capture process, strengthening the case for grouping acquisition-related views. [17]

Other dataset details that remain unspecified are the exact Open Images food-label/crop selection rule, bounding-box expansion policy, duplicate-detection method, geographic and cuisine composition, per-task

class distribution, annotation-review procedure for the 302,157-record compilation, and the rationale for selecting 3,490 Nutrition5K examples rather than the full source dataset [1]. These variables belong in the final data statement.

Data collection and preprocessing. Before any full GPU run, the dataset should pass the following ordered process [1]:

First, **integrity verification** resolves the exact input path and checks archive SHA-256, CRC integrity, expected archive members, manifest checksums, and expected record counts. Second, **schema verification** checks uniqueness of sample IDs, required conversation turns, matching image placeholders and media locations, required answer tags, parseable structured labels, and existence of every referenced media object. Third, **leakage verification** asserts that no group identifier crosses train, validation, or test boundaries. Fourth, **image verification** decodes a representative, source-stratified set as RGB and records corrupt or unsupported media. Fifth, **model preprocessing** applies the pinned Food-R1/Qwen3-VL processor within the specified pixel budget of 65,536 to 589,824 pixels per image. The blueprint does not specify independent RGB normalization constants, so preprocessing should be inherited from the frozen model processor rather than invented. Sixth, **text processing** uses the pinned tokenizer and chat template, enforces a maximum sequence length of 2,048 tokens, verifies image-token alignment, and masks labels according to the training implementation. Seventh, **structured target processing** canonicalizes field names and units before metric computation without changing semantic labels.

Feature engineering. Because Food-R1 is a vision-language model, ThriveLens does not require hand-crafted visual descriptors such as color histograms or SIFT features. The principal feature-engineering decisions operate at the representation contract:

Feature-engineering layer	Implementation	Purpose
Visual representation	Food-R1/Qwen3-VL processor with project pixel bounds	Preserve pretrained visual representation while bounding memory
Task representation	Stable instruction templates and task IDs	Distinguish classification, ingredients, nutrition, and structured tasks
Output schema	Canonical JSON keys and normalized units	Permit deterministic parsing and metric computation
Group metadata	Original/acquisition/source group identifier	Prevent leakage of related images
Source metadata	Open Images, Fruits-360, Nutrition5K, reconciled additional lineage	Enable source-stratified metrics
Nutrient representation	Canonical units plus provider/ nutrient-basis identifiers	Separate measured/catalogue values from model estimates
Provenance	Dataset, model, processor, adapter, request, provider revisions	Reconstruct the origin of every result

Feature-engineering layer	Implementation	Purpose
Application features	Confirmed food ID, portion, correction flag, uncertainty state	Measure the human-confirmed pipeline independently of raw model output

Candidate models. The experimental comparison should prioritize hypotheses and architectural relevance rather than maximize the number of models.

Candidate	Role	Justification	Publication condition
Unadapted Food-R1	Mandatory primary baseline	Measures whether ThriveLens-specific QLoRA adds value over the released food-specialized checkpoint	Run on exactly the same frozen test manifest
Food-R1 plus QLoRA rank 64	Primary proposed model	Matches the blueprint's current Cell 02 configuration	Full three-epoch run and untouched held-out evaluation required
Food-R1 plus QLoRA rank 32	Adapter-capacity ablation	Tests whether lower adapter rank delivers similar quality at lower trainable capacity	Same data, schedule, seeds, processor, and evaluation
Food-R1 with limited vision adaptation	Optional architecture ablation	Tests whether freezing the vision stack constrains domain adaptation	Must receive explicit compute budget; otherwise omit
Qwen3-VL-8B base	Optional upstream baseline	Quantifies gains inherited from Food-R1 before ThriveLens adaptation	Useful only if evaluated with exactly the same task prompts
MobileNetV2	Legacy visual classifier baseline	Provides lightweight conventional comparison for compatible classification tasks	Must use compatible labels/splits; cannot be compared directly on generative nutrition tasks
Full 302,157 vs. balanced 96,000 manifest	Data-scale ablation	Tests whether full heterogeneous data improve robustness over the earlier budget configuration	Keep architecture and tuning budget controlled

MobileNetV2 cannot substitute for Food-R1 on ingredient generation, structured nutritional output, or multimodal dialogue. Its comparison should therefore be restricted to a shared food-classification endpoint. MobileNetV2 was designed as an efficient mobile architecture, which makes it meaningful as an efficiency baseline, but not as a functionally equivalent VLM. [28]

Primary training protocol. The blueprint's recorded QLoRA configuration is reproduced without silent modification:

Parameter group	Fixed ThriveLens configuration
Base checkpoint	`zy12123/Food-R1`
Training method	Supervised QLoRA
Epochs	3
Early stopping	Disabled
Best-model replacement	Disabled
GPUs	2 × NVIDIA T4
Micro-batch	1 example per GPU
Gradient accumulation	8
Effective global batch	16
Optimizer	Paged AdamW, 8-bit
Learning rate	2×10^{-4}
Schedule	Cosine
Warmup ratio	0.05
Weight decay	0.01
Adam betas	0.9, 0.999
Adam epsilon	10^{-8}
Maximum gradient norm	0.3
Base quantization	4-bit NF4
Double quantization	Enabled
Compute precision	FP16
LoRA rank	64
LoRA alpha	16
LoRA dropout	0.05
LoRA targets	Language-side `all-linear`, with specified exclusions
Vision tower and merger	Frozen
LoRA bias	None
DoRA	Disabled
Maximum sequence length	2,048

Parameter group	Fixed ThriveLens configuration
Minimum image pixels	65,536
Maximum image pixels	589,824
Loader workers	4
Prefetch factor	2
Pinned memory	Enabled
Persistent workers	Enabled
Distributed optimization	DeepSpeed ZeRO-2 and NCCL
Gradient checkpointing	Enabled with non-reentrant behavior
Checkpoint interval	Every 1,000 optimizer steps
Retained checkpoints	4
Logging interval	Every 10 steps
Validation	Full validation after each epoch
Final generation	Entire test set when `MAX_TEST_SAMPLES=0`
Generation limit	512 new tokens
Decoding	Greedy, no sampling, one beam
Telemetry	Every 60 seconds

The configuration's NF4, double-quantization, and paged-optimizer choices follow the principal memory-saving components introduced by QLoRA. [12] LoRA rank 64 represents a project engineering choice rather than a value prescribed by the original LoRA paper. [11] Likewise, the ThriveLens learning rate should not be confused with the original Food-R1 authors' 4×10^{-5} full-fine-tuning learning rate. [5]

Hyperparameter tuning strategy. The fixed three-epoch acceptance run remains unchanged, but candidate selection can be performed before it. A computationally defensible tuning space is:

Hyperparameter	Baseline/ recorded	Proposed comparison values	Selection data
LoRA rank	64	32, 64	Training-only screening plus project validation
Learning rate	2×10^{-4}	1×10^{-4}, 2×10^{-4}, 3×10^{-4}	Validation
LoRA dropout	0.05	0.00, 0.05, 0.10	Validation
LoRA alpha	16	16, 32	Validation

Hyperparameter	Baseline/ recorded	Proposed comparison values	Selection data
Maximum visual budget	589,824 pixels	lower budget vs. recorded maximum	Validation plus GPU-memory/ throughput
Data manifest	302,157	96,000 balanced vs. 302,157 full	Controlled ablation
Vision adaptation	Frozen	frozen vs. explicitly limited adaptation	Controlled ablation

Only rank 32 versus rank 64 and full-data versus 96,000-record comparisons are explicitly identified by the blueprint; the additional learning-rate, dropout, alpha, and visual-budget values above are **proposed tuning options**, not previously executed project settings. They should be reduced if the fixed compute allocation cannot support a clean comparison.

A single scalar multi-task selection criterion is **unspecified** in the blueprint. Before tuning, the experiment should therefore preregister either a clearly identified primary task or a task-macro composite. A defensible composite converts each validation metric to a directionally consistent normalized score and then macro-averages by task rather than allowing a numerically large metric such as calorie error to dominate smaller-scale F1 scores. Raw per-task results must remain visible even when a composite is used.

Run reproducibility. Every training run must record the Git commit, notebook hash, archive hash, manifest hash, split counts, source and task counts, base-model revision, processor/tokenizer revision, adapter configuration, dependency versions, GPU identity, CUDA and PyTorch versions, random seed, optimizer and scheduler parameters, image and token limits, start/end UTC timestamps, completed optimizer steps, checkpoint hashes, validation history, held-out command, and failure summary [1]. This follows the broader reproducibility principle that code, data, experimental configuration, and reporting should make published machine-learning findings independently inspectable. [29]

The exact random seed, operating-system image, CUDA version, PyTorch version, Transformers version, PEFT version, bitsandbytes version, and DeepSpeed version of the **successful final run are currently unspecified**. They should not be guessed in the paper; they must be copied from the successful runtime manifest.

Experimental Design and Results

Validation and cross-validation strategy. The final 52,941-record test split must remain untouched until all architecture and hyperparameter decisions have been made [1]. Using test outcomes to choose rank, learning rate, checkpoint, prompt, parsing rules, or confidence thresholds would convert the held-out set into tuning data and produce optimistic estimates.

The recommended hierarchy is:

Training phase: Fit candidate models on the 242,756-record training partition only.

Hyperparameter-screening phase: Where compute permits, conduct three-fold group-aware cross-validation **within the training partition**, preferably using the 96,000-record balanced budget manifest for preliminary comparisons. `StratifiedGroupKFold` is appropriate when both class balance and non-overlapping related-image groups matter. [30] Where task labels are not naturally stratifiable, `GroupKFold` still ensures that a group does not occur in both sides of a fold. [31] The exact group identifier must be fixed before this stage.

Validation phase: Train the most promising candidate configurations on the complete project training split and compare them on the 6,460-record validation split. Confidence-temperature fitting and abstention thresholds are also determined here, not on test data.

Test phase: Freeze model, processor, parser, calibration transformation, and thresholds; then run deterministic generation once across all 52,941 test records and compute the preregistered metrics.

Full nested cross-validation is not economically sensible for every 8-billion-parameter VLM run under the project's dual-T4 constraints. It is, however, appropriate for the much cheaper MobileNetV2 comparison or other conventional baselines because nested CV estimates the combined error of hyperparameter selection and final fitting without using the same validation observations to advertise final performance. [32]

Evaluation metrics. No single metric captures the ThriveLens objective. The evaluation therefore separates parsing, semantic correctness, numerical accuracy, uncertainty, system integrity, and human interaction.

For a target y_i, prediction $\hat{y}_i$, and n examples, mean absolute error is

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|, \tag{2}$$

root mean squared error is

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}, \tag{3}$$

and mean absolute percentage error is

$$MAPE = \frac{100}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right|, \tag{4}$$

with zero or near-zero reference values handled by a preregistered rule rather than silently excluded.

For ingredient sets,

$$Precision = \frac{TP}{TP + FP}, \quad Recall = \frac{TP}{TP + FN}, \tag{5}$$

and

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}. \tag{6}$$

The full metric matrix is:

Evaluation dimension	Metrics
Structured response	JSON/schema validity, required-field completion, invalid-generation rate
Exact structured prediction	Exact match, field-level exact match
Text/token tasks	Token precision, recall, F1
Ingredients	Set precision, recall, F1, optional IoU
Calories/nutrients/ portion	MAE, RMSE, MAPE or percentage MAE, signed bias
Classification	Top-1 accuracy, macro F1, source-stratified accuracy
Calibration	Expected calibration error, reliability plot, Brier-style score where probabilistic output exists
Abstention	Coverage, error-at-coverage, abstention precision, unresolved rate
Robustness	Metrics by source, task, image-resolution bucket, cuisine/category where metadata permit
Human confirmation	Correction rate, confirmed-food accuracy, portion edit rate, completion time
Nutrition pipeline	Error before confirmation vs. after catalogue confirmation, missing-provider rate
Offline persistence	Restart survival, offline mutation survival, recovery rate
Synchronization	Duplicate writes, conflicts, successful retries, version violations
Security	Cross-account reads/writes blocked, expired-session rejection, RLS policy coverage
Serving	Latency, throughput, timeout rate, invalid-response rate, model-revision presence
Device	Camera, barcode, location, notification, resume, and OAuth success
Coaching	Groundedness, record-citation validity, invalid action rejection, approval rate

Calibration requires explicit analysis because high task accuracy does not ensure reliable confidence. Temperature scaling can be fit on the validation partition and then frozen before test evaluation. [19]

For the final paper, uncertainty around principal metrics should be reported using group-aware or source-stratified bootstrap confidence intervals. Paired bootstrap resampling is preferred for two models evaluated on the same held-out records because it estimates the distribution of their per-example performance difference while preserving pairing. Related views belonging to one underlying acquisition group should be resampled together rather than independently.

Experimental matrix.

Experiment	Question tested	Controlled comparison	Primary outputs
E0	Released baseline quality	Unadapted Food-R1	Full project held-out metrics
E1	Main ThriveLens hypothesis	Food-R1 QLoRA rank 64 vs. E0	Structured metrics, numeric errors, calibration
E2	Adapter capacity	Rank 32 vs. rank 64	Performance, trainable parameters, memory, throughput
E3	Visual adaptation	Frozen vision vs. limited vision adaptation	Domain robustness vs. memory/compute
E4	Data scale	96,000 balanced vs. 302,157 full	Overall and source-balanced generalization
E5	Nutrition architecture	Direct model nutrients vs. human-confirmed catalogue calculation	Nutrient error, correction rate, completion time
E6	Offline resilience	Online path vs. forced offline/ reconnect conditions	Data survival, duplicate count, derived-total correctness
E7	Security	Owner access vs. cross-account access	RLS pass/fail matrix
E8	Coach safety	Free-form proposal vs. typed grounded action	Unsupported claims and invalid-action rate

Dataset audit results. The split arithmetic provides the first measured result available from the blueprint: 242,756 + 6,460 + 52,941 = 302,157. By contrast, source arithmetic yields 107,794 + 183,893 + 3,490 = 295,177, a difference of 6,980 records [1]. Thus, 97.69% of the declared dataset currently has source attribution through the three reported counts and 2.31% remains unreconciled.

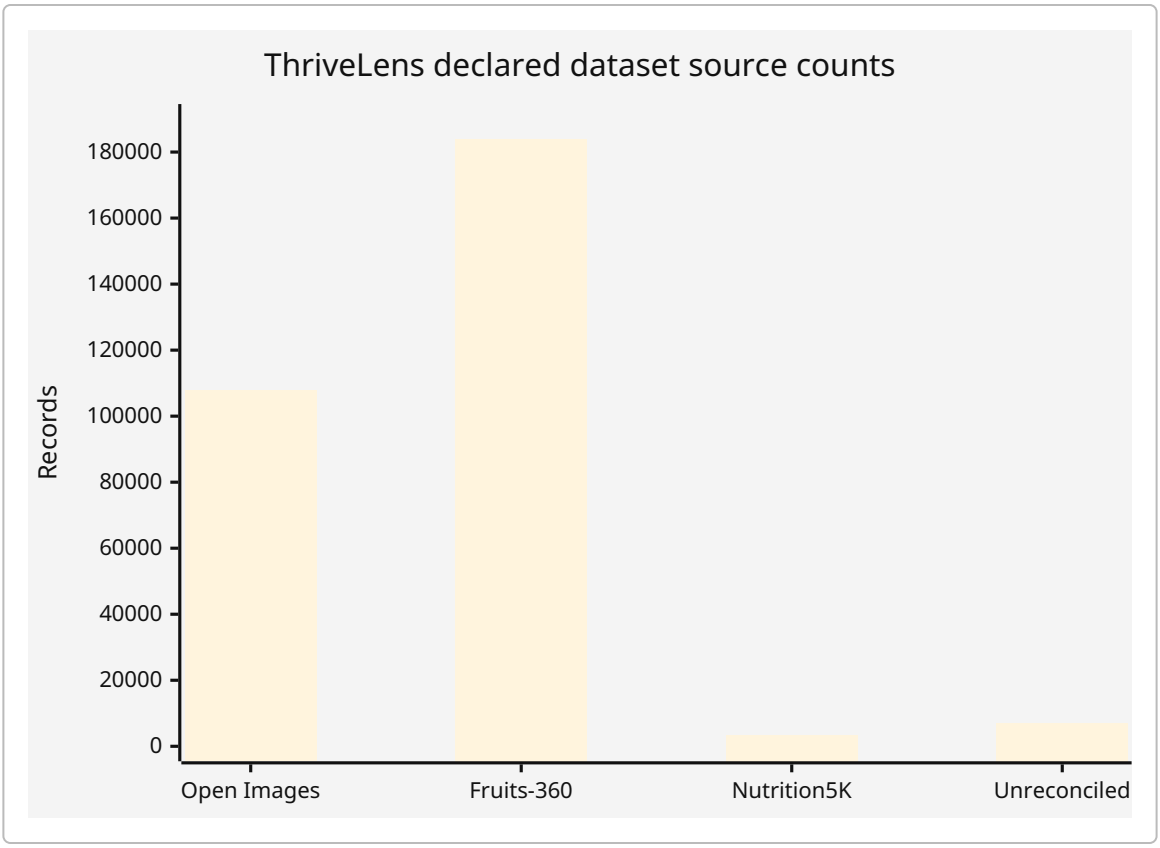


Fig. 2. Dataset-source audit derived from the supplied ThriveLens blueprint. The 6,980-record bar represents an arithmetic lineage gap, not a newly identified data source [1].

The composition also indicates an important domain imbalance: 60.86% of the total declared records are attributed to Fruits-360, while Nutrition5K contributes only 1.16%. Since Fruits-360 includes repeated controlled object imagery whereas Nutrition5K emphasizes real composed dishes with measured nutritional attributes, aggregate accuracy could be dominated by an easier visual domain unless results are macro-averaged or stratified by source. The contrasting source characteristics are documented by the respective official dataset and research sources. [33]

Repository audit results. The mobile code baseline is also measurable without inferring model performance. The repository states that all 15 application screens are implemented at the UI level but that AI responses, authentication, and food scanning are simulated and that the machine-learning model is not integrated. The package manifest confirms SQLite, Supabase, and Drizzle dependencies but cannot by itself establish that authenticated persistence and synchronization work. The SQL file explicitly disables RLS, making secure two-account isolation unproven at the audited baseline. This conflicts with Supabase's current production guidance that RLS should protect exposed tables. [22]

The legacy MobileNetV2 baseline also requires correction before publication. The actual pinned script differs from the blueprint narrative and should be reported according to the repository code: random 80/20 directory split, seed 123, ImageNet MobileNetV2, frozen first-stage backbone, global-average pooling, dropout 0.2, softmax classification, initial Adam optimization, a subsequent two-epoch last-30-layer fine-

tuning stage at 10^{-5}, and optional TFLite export. No result should be attributed to the alternative augmentation-and-callback configuration unless that configuration is committed and executed separately.

Published Food-R1 context. The following numbers come from the original Food-R1 study and serve only as external baseline context. They are not outcomes of the ThriveLens 302,157-record QLoRA experiment.

Published model	CalorieBench-80K MAE	RMSE	R^2
Qwen3-VL-8B	195.18	370.04	-1.13
GPT-4o-mini	88.40	167.00	0.57
FoodLMM	170.35	274.22	-0.19
FoodLMM-s1	57.99	134.46	0.71
Food-R1	**42.56**	**112.89**	**0.81**

These published values are reproduced from the Food-R1 investigators' CalorieBench-80K evaluation. [5]

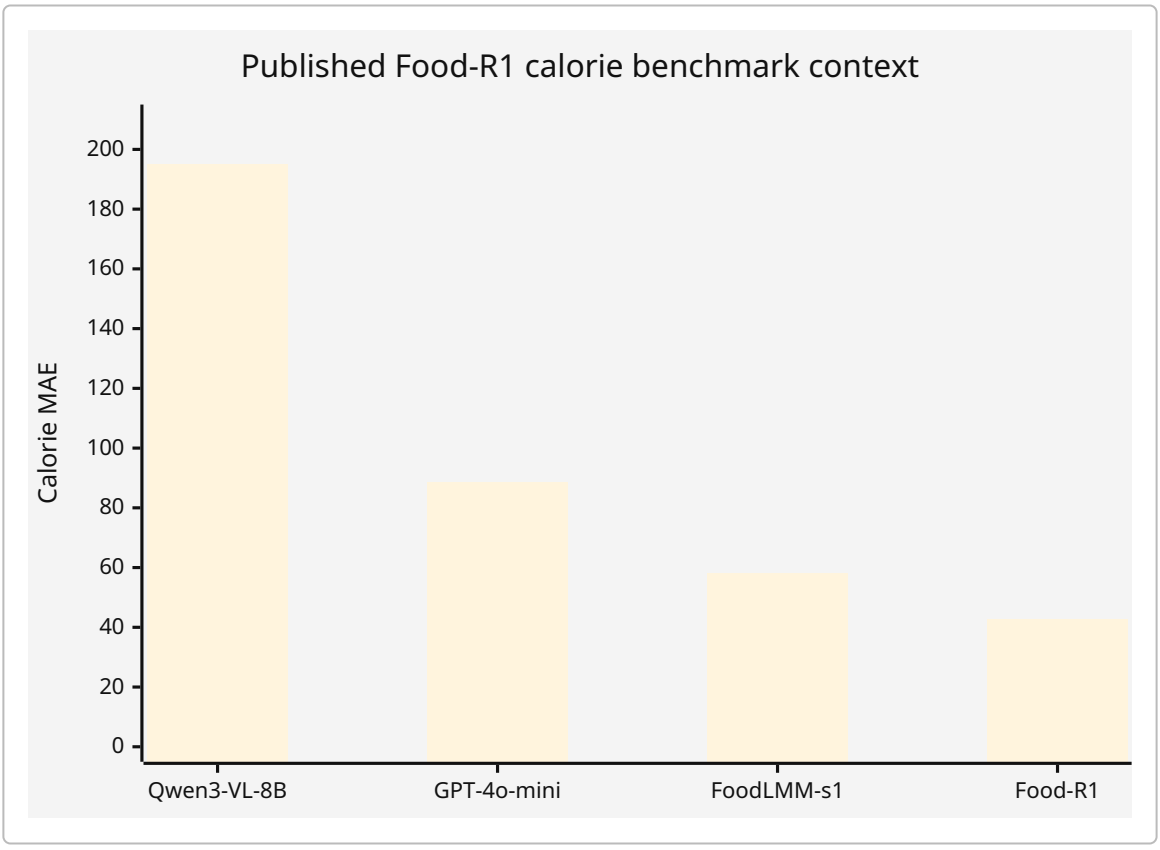


Fig. 3. Published CalorieBench-80K MAE; lower values are better. This figure describes the original Food-R1 publication, not the ThriveLens QLoRA adaptation. [5]

The same Food-R1 paper reports the following additional external context:

Dataset/task	Published Food-R1 result
Food-101 classification	95.21% top-1 accuracy
VireoFood-172 ingredient recognition	IoU 78.50
VireoFood-172 ingredient recognition	F1 84.61
Nutrition5K calorie estimation	MAE 27.6
Nutrition5K calorie estimation	percentage MAE 14.4%
Nutrition5K fat estimation	MAE 3.2
Nutrition5K carbohydrate estimation	MAE 4.4
Nutrition5K protein estimation	MAE 4.0

[5]

These values provide a sanity check for a future ThriveLens run. They must not be used to populate ThriveLens project result cells because the underlying dataset, training objective, post-training procedure, and evaluation manifests differ.

ThriveLens project result sheet. The scientifically correct current project table is therefore:

Outcome	Unadapted Food-R1	ThriveLens QLoRA rank 64	Rank 32 ablation	Evidence status
JSON validity	Not yet measured	Not yet measured	Not yet measured	Requires held-out run
Exact match	Not yet measured	Not yet measured	Not yet measured	Requires held-out run
Token F1	Not yet measured	Not yet measured	Not yet measured	Requires held-out run
Ingredient F1	Not yet measured	Not yet measured	Not yet measured	Requires held-out run
Calorie MAE	Not yet measured	Not yet measured	Not yet measured	Requires held-out run
Nutrient RMSE	Not yet measured	Not yet measured	Not yet measured	Requires held-out run
MAPE	Not yet measured	Not yet measured	Not yet measured	Requires held-out run
Calibration error	Not yet measured	Not yet measured	Not yet measured	Requires calibrated evaluation

Outcome	Unadapted Food-R1	ThriveLens QLoRA rank 64	Rank 32 ablation	Evidence status
Abstention coverage	Not yet measured	Not yet measured	Not yet measured	Threshold must be fixed on validation
Test records processed	Not yet measured	Not yet measured	Not yet measured	Target = 52,941
GPU peak memory	Not yet measured	Not yet measured	Not yet measured	Runtime telemetry
Training throughput	Not applicable	Not yet measured	Not yet measured	Runtime telemetry
Serving latency	Not yet measured	Not yet measured	Not yet measured	Serving benchmark

The absence of values is itself an important evidence result: the blueprint states that the notebook had no saved execution outputs at audit time and explicitly prohibits presenting unexecuted measurements as achieved performance [1].

Human-confirmation experiment. RQ2 should be tested with a paired design. Each sampled meal image is processed through two pathways using the same image and reference record. Path A records the model's direct food/nutrient output. Path B presents candidate foods, collects user confirmation or correction, resolves an Open Food Facts or USDA FoodData Central catalogue record, confirms a portion, and performs deterministic nutrient arithmetic. The primary outcome is the within-example reduction in absolute nutrient error:

$$\Delta MAE = MAE_{\text{direct model}} - MAE_{\text{confirmed pipeline}}. \tag{7}$$

A positive value indicates improvement from human confirmation and catalogue grounding. Completion time, number of edits, catalogue failure rate, and abstention rate are secondary outcomes. The final study sample size is **unspecified** in the blueprint. A power calculation should therefore be performed after a pilot estimates the variance of paired nutrient-error differences rather than inventing a participant count.

Offline and synchronization experiment. RQ3 uses deterministic fixtures. A test account creates, edits, and deletes records online and offline; the application is terminated; the network is restored; identical queued operations are deliberately replayed; and a second account attempts to access the first account's records. A successful implementation must preserve every valid offline mutation exactly once, restore in-progress workout state after restart, maintain correct daily nutrition/workout totals, isolate account data, and make conflict outcomes explicit [1]. Expo's guarantee that its SQLite database persists across application restarts supports the local storage premise, although synchronization correctness remains a ThriveLens implementation property that must be tested. [20]

Security experiment. For every user-owned table and storage path, positive tests verify that the owner can perform authorized operations and negative tests verify that another authenticated user and an anonymous session cannot. Supabase recommends separate RLS policies for SELECT, INSERT, UPDATE, and DELETE, with `auth.uid()` used to bind access to the current authenticated user, and provides database-

testing workflows for RLS. [22] A release result should therefore report both RLS policy coverage and zero unauthorized success cases in the acceptance suite, rather than simply stating that RLS is enabled.

Proposed execution timeline. Formal project milestone dates are unspecified in the blueprint. The following eight-week schedule is therefore an explicit planning assumption beginning after the September 2026 audit, not a claim about already completed work.

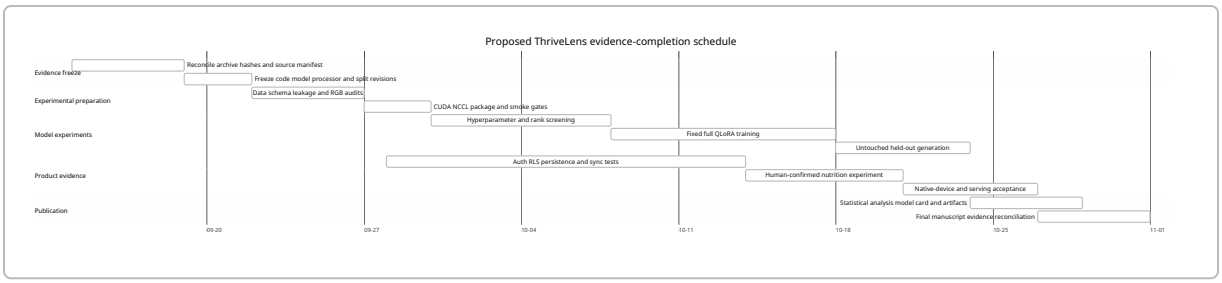


Fig. 4. Proposed execution schedule. The dates are assumed planning dates because the supplied blueprint defines release gates but does not prescribe a publication calendar [1].

Discussion, Limitations, and Ethical Considerations

Interpretation of the proposed contribution. ThriveLens should be evaluated primarily as an architectural separation between **observation, confirmation, evidence, computation, and action**. A conventional food classifier answers a narrower question: which class most resembles an image? Food-R1 broadens the interface to ingredients, food descriptions, structured nutritional fields, and reasoning. [5] That breadth creates utility, but it also creates additional failure modes because a fluent structured prediction can appear authoritative even when image evidence is insufficient.

The human-confirmed pipeline directly addresses this risk. Food-R1 identifies candidates; the application exposes uncertainty; a person confirms what was consumed and corrects portion information; a nutrition provider supplies identifiable source data; nutrient totals are recalculated deterministically; and only then is the meal committed [1]. The result is not an assumption that human input is perfect. Rather, it creates auditable boundaries between different sources of error. A future error can be traced to recognition, user confirmation, provider data, portion choice, or arithmetic rather than being buried inside one generative output.

Expected effect of QLoRA. QLoRA is technically appropriate for the project's compute constraints because it avoids full-model updates while preserving a high-capacity pretrained backbone. [12] Nevertheless, H1 should not be treated as self-evident. Food-R1 has already been specialized for food tasks using multi-task supervised learning and reinforcement fine-tuning. [5] Additional supervised adaptation can improve domain alignment, but it can also cause regressions, distribution overfitting, forgetting, schema specialization, or deterioration of capabilities not sufficiently represented in the 302,157-record manifest. The unadapted Food-R1 baseline is consequently mandatory.

The rank-32 versus rank-64 comparison is especially valuable because higher adapter capacity may improve fitting while increasing optimization state and potentially overfitting dominant sources. A statistically

insignificant improvement at rank 64 combined with greater memory or training time would justify the smaller adapter; a consistent source-balanced advantage would justify rank 64.

Dataset imbalance and external validity. The strongest dataset concern is not total sample count but composition. More than 60% of the declared records are assigned to Fruits-360, whereas only 1.16% are assigned to Nutrition5K. Fruits-360 is valuable for fine-grained produce recognition, but its acquisition process involves repeated, comparatively controlled object imagery. [17] Nutrition5K contains substantially more naturalistic composed dishes and nutritional ground truth. [6] A model can therefore obtain a strong aggregate metric by performing exceptionally on a dominant controlled domain while remaining weak on mixed meals—the use case that matters most for nutrition.

The response is source-balanced reporting. The final paper should publish overall results, per-source results, and a macro-source average. Where class/cuisine metadata permit, performance should also be disaggregated by food group, visual complexity, number of ingredients, and geographic or cuisine category. WHO's multimodal-model guidance identifies biased or poor-quality training data and unequal performance as relevant health-AI risks, making subgroup evaluation ethically important as well as statistically informative. [34]

The 6,980-record lineage discrepancy is another publication blocker. It does not necessarily imply corrupt data, but it means the current description is not sufficient to reproduce the dataset exactly. Likewise, the two archive hashes mean two researchers could follow the written protocol and still use different inputs. Reconciliation must occur before the final run rather than afterward.

Portion and nutrient limitations. Food photographs are inherently incomplete evidence for exact nutrient intake. A two-dimensional RGB image may not reveal depth, cooking oils, sugar, sauces, ingredients hidden beneath other food, or preparation-specific changes. Nutrition5K's inclusion of component weights and depth data itself reflects the difficulty of recovering nutritional quantities from ordinary imagery. [6] ThriveLens therefore should not frame a model-generated calorie number as a measurement.

USDA FoodData Central and Open Food Facts also serve different roles. FoodData Central supplies standardized food and nutrient data through official USDA APIs and is released under CC0. [27] Open Food Facts is especially useful for packaged products and barcode workflows but has its own licensing, completeness, contributor, and rate-limit considerations. [35] Provider identifiers, nutrient basis, serving units, data freshness, and missing-field state must therefore be persisted with the meal snapshot.

Calibration limitation. A model's raw confidence field should not automatically be interpreted as a calibrated probability of correctness. Modern neural networks can be miscalibrated even when classification performance is strong. [19] Calibration should be fit only on validation data and evaluated on the untouched test set. Furthermore, a calibrated recognition probability cannot resolve inherently unobservable information such as hidden ingredients. Abstention must therefore consider both model uncertainty and task observability.

Fixed-epoch training limitation. The blueprint requires three epochs and disables early stopping [1]. This has the advantage of making model runs directly comparable under a fixed schedule, but it also prevents an optimizer from terminating when validation quality starts deteriorating. All validation histories must therefore be retained. The final paper should report each epoch's validation results and state whether the

final-epoch checkpoint improved or regressed, even though the fixed protocol does not permit silently substituting a better earlier checkpoint as the nominal final run.

Compute limitation. Two T4 GPUs are a constrained environment for adapting and evaluating an 8-billion-parameter multimodal system. QLoRA and ZeRO are appropriate mitigations, but actual throughput depends on image resolution, sequence lengths, batch composition, data-loader behavior, quantization kernels, checkpointing overhead, and distributed communication. [36] The study must therefore report observed examples per second, optimizer steps per second, GPU utilization, peak allocation, elapsed training time, and held-out generation time rather than derive runtime claims from nominal T4 specifications.

Legacy baseline limitation. The current MobileNetV2 script is not functionally comparable with Food-R1. It is a closed-set image classifier, whereas Food-R1 is a generative multi-task VLM. A fair comparison is possible only on a shared classification target. Furthermore, the script currently creates its own random 80/20 directory split rather than demonstrating use of the same project split. A baseline run should therefore be modified and versioned to consume frozen train/validation/test identifiers before it is included in the final paper.

Application baseline limitation. The current repository demonstrates interfaces rather than complete scientific behavior. Simulated authentication and food/coach responses mean current screenshots cannot validate RQ2, RQ3, or RQ5. Similarly, dependency presence does not prove persistence, and SQL-table presence does not prove account security. The experiment therefore correctly treats native-device, database-security, and end-to-end tests as separate measurement gates.

Privacy and health-data ethics. Fitness and nutrition records may reveal sensitive habits, body measurements, locations, and health-related behaviors. WHO's 2021 AI-for-health guidance argues that ethics and human rights should be central to design, deployment, and use, with attention to autonomy, transparency, responsibility, inclusion, safety, and sustainability. [37] ThriveLens should consequently minimize collected data, obtain informed consent for optional analytics and model-improvement use, protect private images, provide export and deletion functions, clearly distinguish measured from estimated values, and avoid turning engagement incentives into pressure around food or body weight.

The target data architecture should use Supabase Auth for identity, private or access-controlled media storage, user-scoped RLS policies, short-lived signed access where appropriate, and deletion workflows covering cloud records, device caches, pending operations, and retained images [1]. Supabase specifically recommends RLS on user-accessible profile information referencing `auth.users`. [38] The current RLS-disabled SQL state must therefore be described as an audited security defect rather than a final implementation.

Autonomy and human oversight. The confirmation stage should remain genuinely editable. Users must be able to reject the candidate food, search manually, change quantity or units, decline a coach action, and use the application without accepting automated recommendations. A confidence value should never be used to remove the confirmation step for safety-relevant fields unless a separate evaluation justifies that design. WHO's LMM guidance emphasizes human oversight and warns that automation bias can cause users to overlook model errors. [39]

Medical boundary. ThriveLens is a consumer fitness and nutrition assistant, not a diagnostic system, emergency service, dietitian, allergy detector, or medical-prescription tool [1]. This boundary should be reflected not only in disclaimers but in capability design: the model should not infer allergy safety from an image, generate medication recommendations, diagnose conditions, or present exercise progression as clinical rehabilitation advice. Health-critical queries should redirect users toward qualified professionals or emergency resources as appropriate.

Bias and international scope. The project's stated worldwide objective creates a higher validation burden than a single-cuisine product [1]. Food names, ingredients, serving conventions, package languages, and visual presentation vary by region. A model that performs well on common Western, restaurant, or controlled produce imagery may fail on underrepresented cuisines. The final model card should therefore identify the geographical and cuisine coverage that was actually measured rather than claim universal competence.

Prompt-injection and generated-action risk. Food labels, user notes, retrieved text, and third-party provider data should be treated as untrusted data rather than executable instructions. Coaching outputs should follow typed schemas and an allowlist of permissible action types. The application should verify referenced workout, food, meal-plan, and user-record identifiers before offering an action; every write should require explicit approval and an idempotent operation identifier [1]. NIST's AI Risk Management Framework promotes lifecycle-wide management of risks to individuals, organizations, and society, while its generative-AI profile adapts that approach to risks specific to generative systems. [40]

Reproducibility and scientific integrity. Reproducibility is not satisfied by publishing only an adapter file. The final artifact set must make it possible to reconstruct which code, data, model, processor, prompt schema, parser, calibration rule, and hardware/software environment produced each reported number. Pineau et al. emphasize code and data access, explicit experimental details, and reproducibility checklists as mechanisms for improving reliability in machine-learning research. [29] ThriveLens's run-manifest concept is therefore central to the paper rather than merely an implementation convenience.

Overall conclusion. The project is scientifically strongest when framed as a transition from a verified interface prototype to a contract-driven, evidence-preserving consumer system. The Food-R1 model offers a strong multimodal starting point, and QLoRA provides a plausible adaptation route under constrained GPU memory. [41] However, the core research question is broader than whether rank-64 QLoRA lowers one benchmark error. ThriveLens must demonstrate that visual predictions can be converted into user-correctable, source-grounded, deterministically calculated, securely persisted, and reproducibly evaluated records without collapsing model inference into truth.

At the current evidence boundary, the project has a defensible methodology but not defensible ThriveLens performance numbers. The final claim of a completed system requires reconciliation of the dataset manifest and archive hash; successful fixed-schedule training; evaluation of the full untouched test set; calibrated or explicitly abstaining inference; provider-grounded nutrition; secure authenticated RLS; offline synchronization; native-device operation; and end-to-end acceptance evidence [1]. Preserving that boundary is more scientifically valuable than reporting unverified high accuracy.

Acknowledgment, References, and Reproducibility Appendix

Acknowledgment— The project team acknowledges the supervision of Sant Kumar Maurya (E13548) and the authors and maintainers of Food-R1, Qwen3-VL, Open Images, Fruits-360, Nutrition5K, Food-101, USDA FoodData Central, Open Food Facts, Expo, Supabase, and the open-source machine-learning infrastructure on which the proposed study depends. Funding-agency information is unspecified in the supplied project material and should therefore not be asserted.

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Reproducibility appendix.

The reproducibility package should be organized around one immutable experiment identity rather than an informal collection of notebooks. A result is considered reproducible only when the model output can be

mapped back to a precise code revision, dataset manifest, model revision, preprocessing configuration, training run, and evaluator.

Reproducibility item	Required artifact	Current evidence state
Application source	Git repository and immutable commit SHA	**Specified:** `244b4bbe79b4e563d` for audited baseline.
Mobile dependency manifest	Exact `package.json` / lockfile	Baseline `package.js` lock must be frozen.
Training notebook	`ThriveLens_FoodR1_Dual_T4_QLoRA_A_Grade_Verified_302157.ipynb`	Specified in blueprint a successful execution ou
Notebook hash	SHA-256 of exact executed notebook	**Unspecified**
Base model	Food-R1 checkpoint and immutable revision	Model identity specifie revision must be frozer available as Qwen3-VL
Processor/ tokenizer	Immutable processor and tokenizer revisions	**Unspecified final revi**
Training archive	Exact archive file	**Unresolved**
Archive SHA-256	One successful-run hash	**Unresolved:** two hash
Dataset manifest	One record per sample with ID/source/group/task/split/media/label metadata	Required; final frozen r executed evidence
Manifest hash	SHA-256	**Unspecified**
Dataset total	302,157	Specified [1]
Source reconciliation	Source counts summing to 302,157	**Incomplete:** 6,980 rec for
Train split	242,756	Specified [1]
Validation split	6,460	Specified [1]
Test split	52,941	Specified [1]
Group key	Stable leakage-prevention grouping	Required but **exact de**
Task distribution	Counts by task	Required but **unspecif**

Reproducibility item	Required artifact	Current evidence state
Seed	Integer seed used by all stochastic components	Required but **unspecif**
Runtime	Kaggle dual NVIDIA T4	Specified target [1]
Operating-system image	Runtime image/revision	**Unspecified until succ**
CUDA version	Exact runtime version	**Unspecified until succ**
PyTorch version	Exact runtime version	**Unspecified until succ**
Transformers version	Exact runtime version	**Unspecified until succ**
PEFT version	Exact runtime version	**Unspecified until succ**
bitsandbytes version	Exact runtime version	**Unspecified until succ**
DeepSpeed version	Exact runtime version	**Unspecified until succ**
Quantization	4-bit NF4, double quantization, FP16 compute	Specified [1]; method b [12]
Adapter	Rank 64, alpha 16, dropout 0.05, language-side linear targets	Specified [1]
Vision model	Frozen tower and merger	Specified [1]
Optimizer	Paged AdamW 8-bit	Specified [1]
Learning rate	2×10^{-4}	Specified [1]
Scheduler	Cosine with 5% warmup	Specified [1]
Global effective batch	16	Specified [1]
Epochs	3	Specified [1]
Early stopping	Disabled	Specified [1]
Sequence limit	2,048 tokens	Specified [1]
Image range	65,536–589,824 pixels	Specified [1]
Checkpoint period	1,000 optimizer steps, retain four	Specified [1]
Generation	Greedy, maximum 512 new tokens, entire test set	Specified [1]

Reproducibility item	Required artifact	Current evidence state
Validation history	Per-epoch complete validation	Required; **not yet mea**
Held-out predictions	Prediction file covering test manifest	Required; **not yet mea**
Failure file	Invalid JSON, decode failures, generation failures, outliers	Required; **not yet mea**
Calibration artifact	Fitted calibration parameters and validation provenance	Required; **not yet mea**
Model card	Intended use, exclusions, dataset lineage, subgroup gaps, calibration, failures, latency, privacy, rollback	Required; **not yet com**
Serving benchmark	Latency, throughput, memory, timeout behavior, revision metadata	Required; **not yet mea**
RLS test report	Positive and negative account-isolation tests	Required; audited base
Offline test report	Restart, retry, conflict, duplicate-operation, reconnect tests	Required; **not yet mea**
Native-device report	Android/iOS camera, barcode, location, notification, OAuth, restart tests	Required; **not yet mea**
End-to-end acceptance	Scan or select, confirm, catalogue, log, workout, totals, places, reminder, grounded action	Required; **not yet mea**

The minimum **run manifest** for every reported model experiment should contain the following logically complete record:

```
experiment_id
project_git_commit
notebook_sha256
dataset_archive_filename
dataset_archive_bytes
dataset_archive_sha256
dataset_manifest_filename
dataset_manifest_sha256
dataset_record_count
train_record_count
validation_record_count
test_record_count
source_counts
task_counts
grouping_rule
base_model_id
```

```
base_model_revision
processor_revision
tokenizer_revision
adapter_method
lora_rank
lora_alpha
lora_dropout
lora_target_modules
frozen_modules
quantization_type
double_quantization
compute_dtype
optimizer
learning_rate
scheduler
warmup_ratio
weight_decay
adam_beta1
adam_beta2
adam_epsilon
maximum_gradient_norm
micro_batch_per_gpu
gradient_accumulation_steps
world_size
effective_global_batch
maximum_sequence_length
minimum_image_pixels
maximum_image_pixels
epochs_requested
epochs_completed
early_stopping_enabled
gradient_checkpointing
deepspeed_stage
distributed_backend
random_seed
operating_system
python_version
pytorch_version
cuda_version
transformers_version
peft_version
bitsandbytes_version
deepspeed_version
gpu_model
gpu_count
run_start_utc
run_end_utc
optimizer_steps_completed
```

```
checkpoint_identifiers
checkpoint_sha256_values
validation_metrics_by_epoch
final_test_prediction_sha256
final_metrics_file_sha256
failure_file_sha256
calibration_artifact_sha256
serving_artifact_revision
notes_and_failures
```

This manifest should accompany the adapter weights, processor/tokenizer files or immutable revisions, evaluator, frozen test IDs, prediction outputs, metrics output, calibration artifact, plots, representative failure cases, model card, and dependency lock. The requirement is consistent with established machine-learning reproducibility practice, in which code, data, hyperparameters, and execution details are treated as part of the scientific result rather than incidental implementation information. [29]

The final publication evidence should additionally bind the **application result to the model result**. Each inference response should preserve a request identifier and model revision; each confirmed food log should preserve the selected candidate or manual correction, catalogue provider and provider record ID, quantity and unit, nutrient basis, model revision, correction status, and timestamp; each synchronized mutation should preserve an operation identifier; and every user-owned cloud row must remain protected by authenticated authorization policies [1]. Supabase's authentication and RLS mechanisms provide the platform primitives for this identity boundary, but their correctness remains an application-level test requirement. [45]

Under this evidence model, a complete ThriveLens research result is reproducible from **raw manifest to model prediction to confirmed domain record to final reported metric**. Until the two dataset hashes, 6,980-record source-lineage difference, successful GPU run, full held-out evaluation, secure RLS migration, native-device behavior, and end-to-end acceptance artifacts have been resolved and measured, those outcomes remain explicitly unspecified rather than assumed.

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[1] [37] https://www.who.int/publications/i/item/9789240029200
https://www.who.int/publications/i/item/9789240029200

[2] [21] [45] https://supabase.com/docs/guides/auth
https://supabase.com/docs/guides/auth

[3] [11] https://arxiv.org/abs/2106.09685
https://arxiv.org/abs/2106.09685

[4] [30] https://scikit-learn.org/dev/modules/generated/sklearn.model_selection.StratifiedGroupKFold.html
https://scikit-learn.org/dev/modules/generated/sklearn.model_selection.StratifiedGroupKFold.html

[5] [7] [41] https://arxiv.org/abs/2606.04986
https://arxiv.org/abs/2606.04986

[6] https://arxiv.org/abs/2103.03375
https://arxiv.org/abs/2103.03375

8 https://github.com/hustvl/Food-R1
https://github.com/hustvl/Food-R1

9 https://huggingface.co/zy12123/Food-R1
https://huggingface.co/zy12123/Food-R1

10 https://arxiv.org/abs/2511.21631
https://arxiv.org/abs/2511.21631

12 36 https://arxiv.org/abs/2305.14314
https://arxiv.org/abs/2305.14314

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https://doi.org/10.1109%2FCVPR.2018.00474

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https://mlanthology.org/eccv/2014/bossard2014eccv-food/

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https://supabase.com/docs/guides/database/postgres/row-level-security

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24 35 42 https://openfoodfacts.github.io/openfoodfacts-server/api/
https://openfoodfacts.github.io/openfoodfacts-server/api/

25 https://openfoodfacts.github.io/documentation/docs/Product-Opener/api/tutorials/license-be-on-the-legal-side/
https://openfoodfacts.github.io/documentation/docs/Product-Opener/api/tutorials/license-be-on-the-legal-side/

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https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.GroupKFold.html

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https://scikit-learn.org/stable/auto_examples/model_selection/plot_nested_cross_validation_iris.html

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